so take $u = x$ then $du = dx$

$dv = \cos x\,dx \Rightarrow v = \sin x$

Applying Integration by parts, we get

$$\int u dv = uv - \int v du$$

$$\Rightarrow \int x\cos x\,dx = x\sin x - \int \sin x\,dx$$

$$\Rightarrow \int x\cos x\,dx = x\sin x + \cos x + c$$

(iii) Let $I = \int \log x\,dx$

Take $u = \log x$ then $du = \frac{1}{x}dx$

$dv = dx \Rightarrow v = x$

Applying Integration by parts, we get

$$\int u dv = uv - \int v du$$

$$\Rightarrow \int \log x\,dx = x\log x - \int x\frac{1}{x}dx$$

$$\Rightarrow \int \log x\,dx = x\log x - x + c$$

(iv) Let $I = \int \sin^{-1} x\,dx$

$u = \sin^{-1}(x), dv = dx$

Then $du = \frac{1}{\sqrt{1-x^2}}, v = x$

$$\int \sin^{-1} x dx = x\sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}}dx$$

$$\int \sin^{-1} x dx = x\sin^{-1} x + \frac{1}{2}\int \frac{dt}{\sqrt{t}}, \text{ where } t = 1-x^2$$

$$= x\sin^{-1} x + \sqrt{t} + c$$

$$= x\sin^{-1} x + \sqrt{1-x^2} + c$$

Example 11.34

Evaluate : $\int \tan^{-1}\left(\frac{2x}{1-x^2}\right) dx$

Solution

Let $I = \int \tan^{-1}\left(\frac{2x}{1-x^2}\right) dx$

Putting $x = \tan\theta \Rightarrow dx = \sec^2\theta\,d\theta$

Therefore, $I = \int \tan^{-1}\left(\frac{2\tan\theta}{1-\tan^2\theta}\right)\sec^2\theta\,d\theta$

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